

Churn Analysis for Fit.Ly

Introduction

Fit.ly is a subscription-based fitness app offering workout articles, instructional videos, workout tracking, and the ability to share workouts. Over the past two quarters, Fit.ly has seen customer churn rise, putting pressure on acquisition costs and on the Product and Marketing teams. I was asked to identify what's driving churn and provide practical recommendations leadership can act on next quarter.

I was provided three datasets for this analysis:

1. Account information (400 customers)
2. Customer support tickets
3. User activity logs

All three were validated and cleaned before analysis to ensure consistent identifiers, correct data types, and reliable values — detailed below.

Data Validation

Every column across the three sources was checked for missing values, duplicates, and consistency with what Product and Engineering described.

account_info

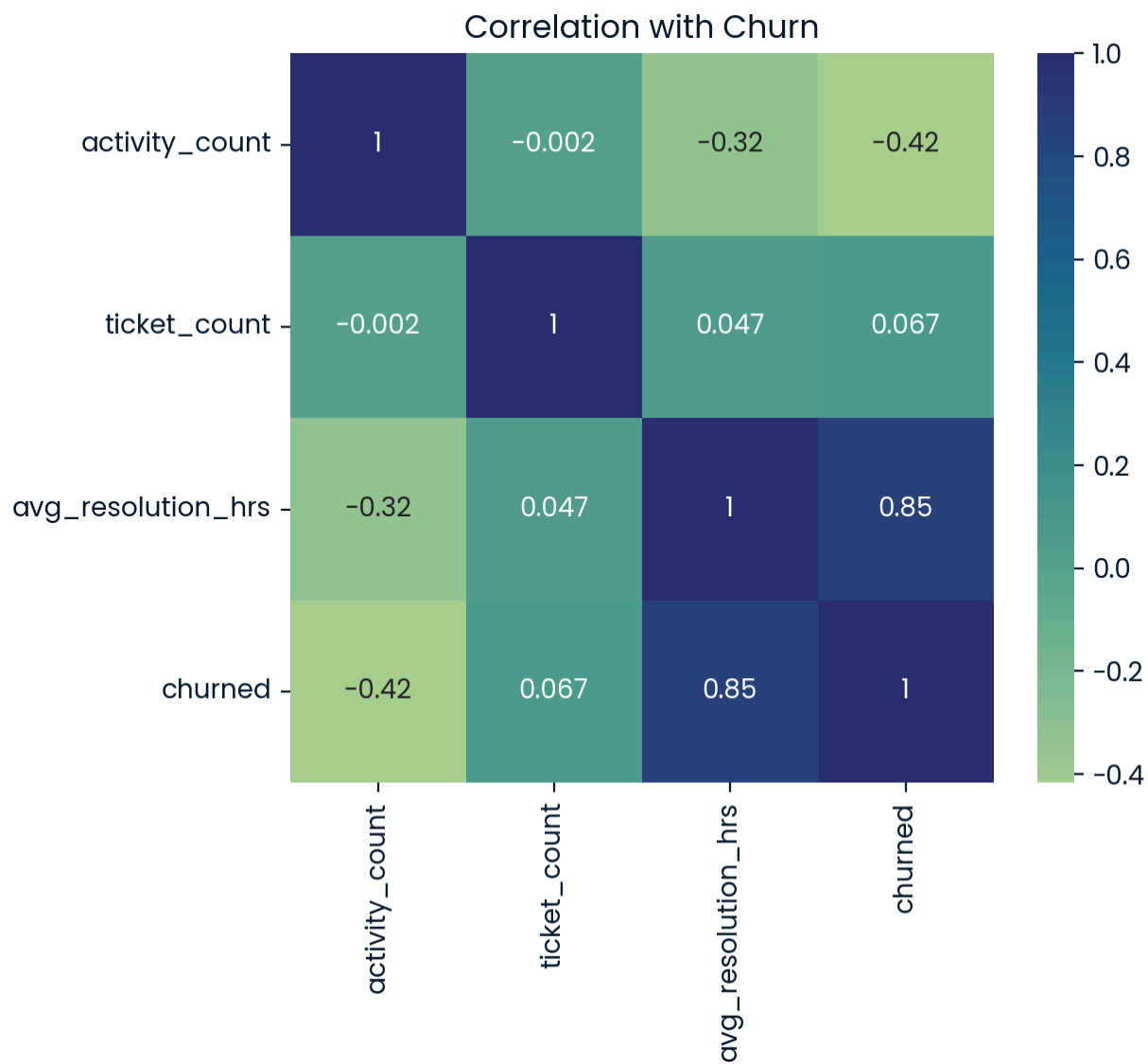
- `customer_id`, `email` — 0% missing, 0 duplicates
- `state` — 0% missing, 50 valid US state names
- `plan` — 0% missing, 4 clean tiers (Free/Basic/Pro/Enterprise)
- `plan_list_price` — 0% missing, consistent non-overlapping ranges per tier
- `churn_status` — 71.5% blank / 28.5% "Y". Per Product, blanks mean active (not missing) — recoded to boolean `churned`
- `signup_date` — described by Product but **does not exist** in the file provided; flagged for Engineering to confirm

customer_support

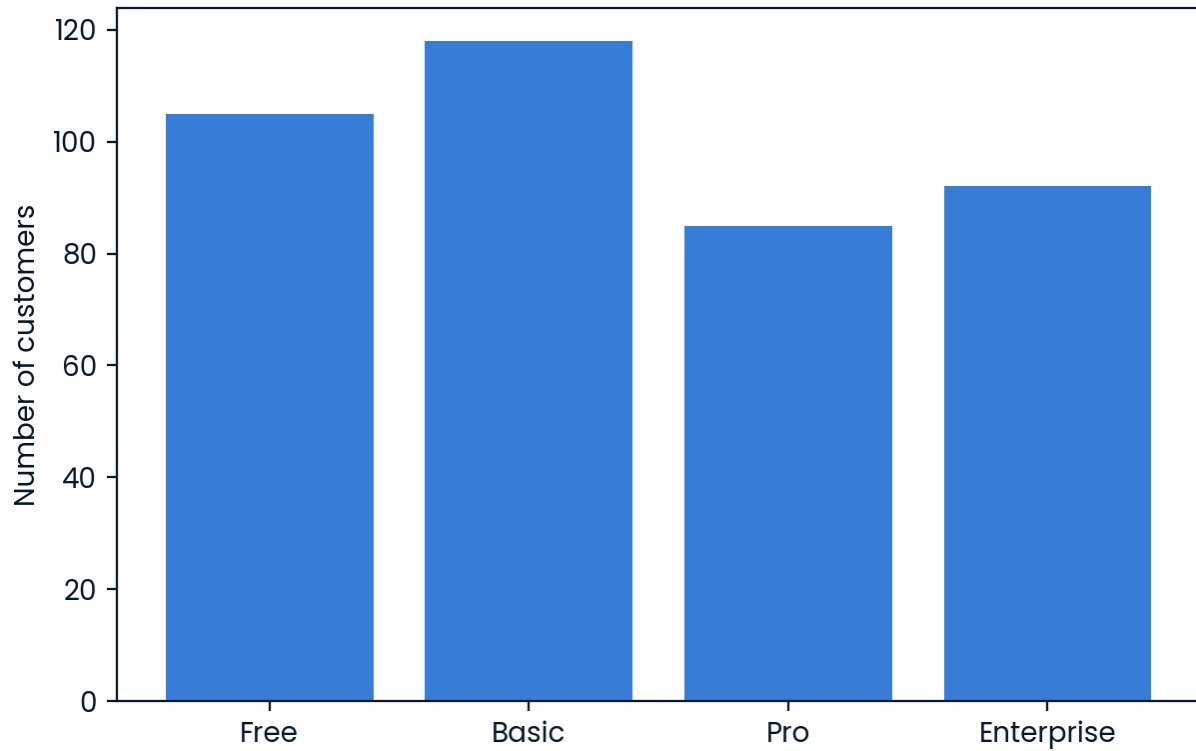
- `ticket_time`, `user_id` — 0% missing, all `user_ids` match `account_info`
- `channel` — 4.2% contain "-" instead of chat/phone/email; treated as "Unknown" rather than dropped
- `topic`, `resolution_time_hours` — clean, no missing or invalid values
- `state` — only contains 0/1, undocumented, unrelated to churn (30.4% either way) — **excluded from analysis**, flagged for the source team
- `comments` — 95% blank as expected; 36 non-blank rows are GDPR data-deletion requests, routed separately as a compliance item

user_activity

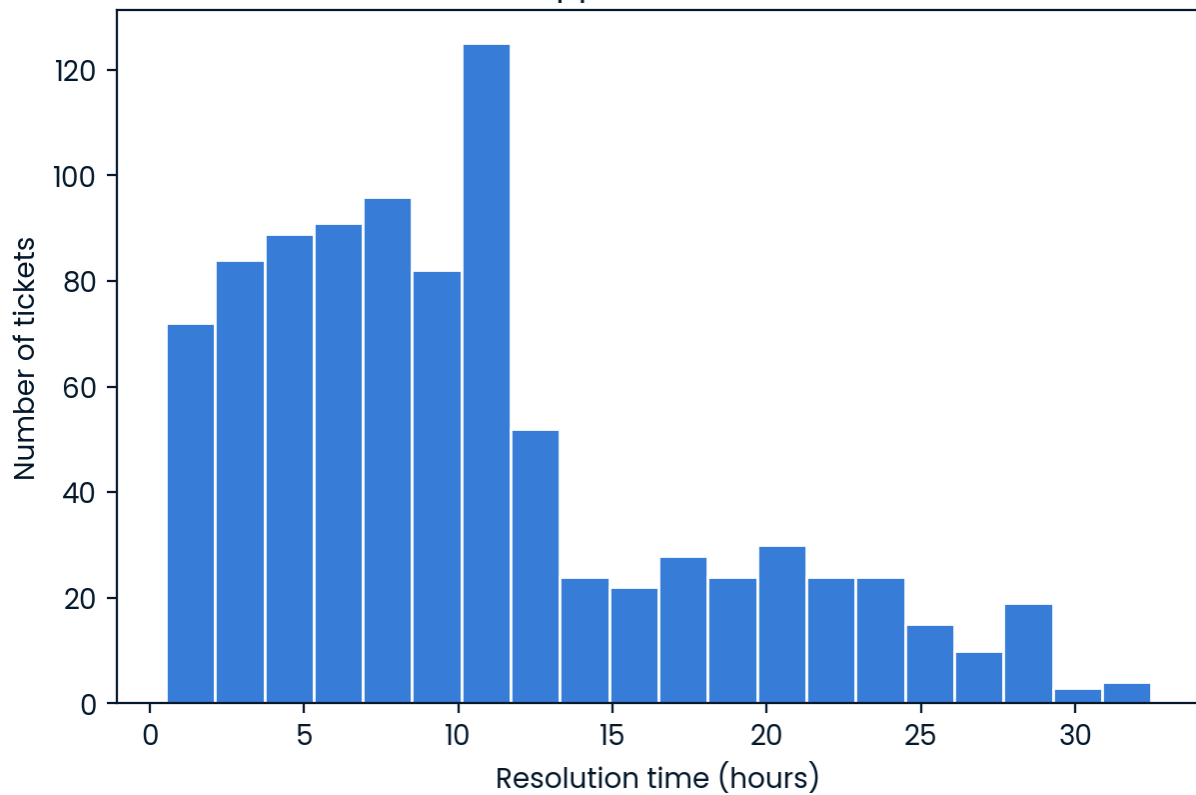
- `event_time`, `user_id`, `event_type` — 0% missing, all `user_ids` match `account_info`
- 154 of 400 customers (38.5%) have zero logged activity — a real signal, not a data gap

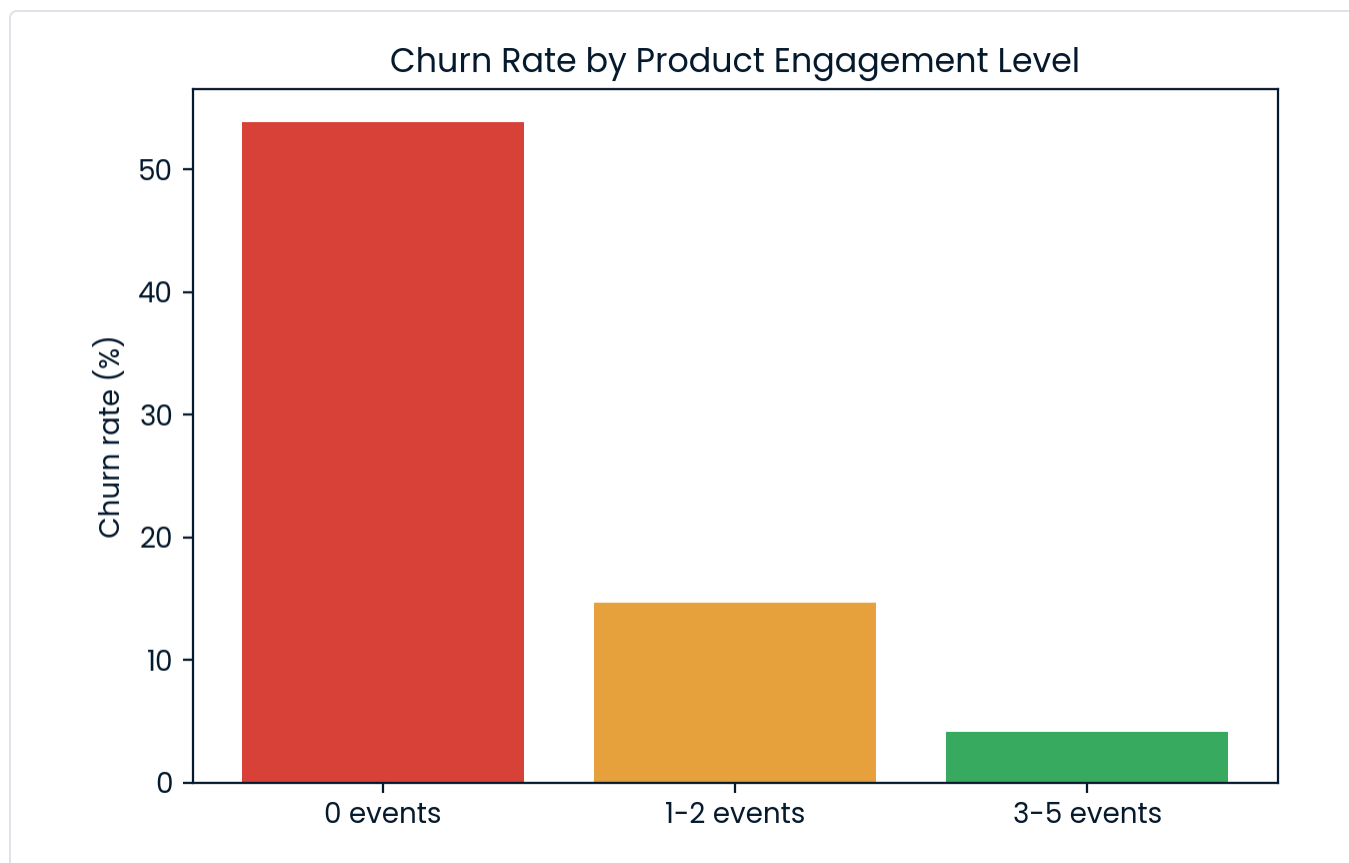


Customers by Plan Tier

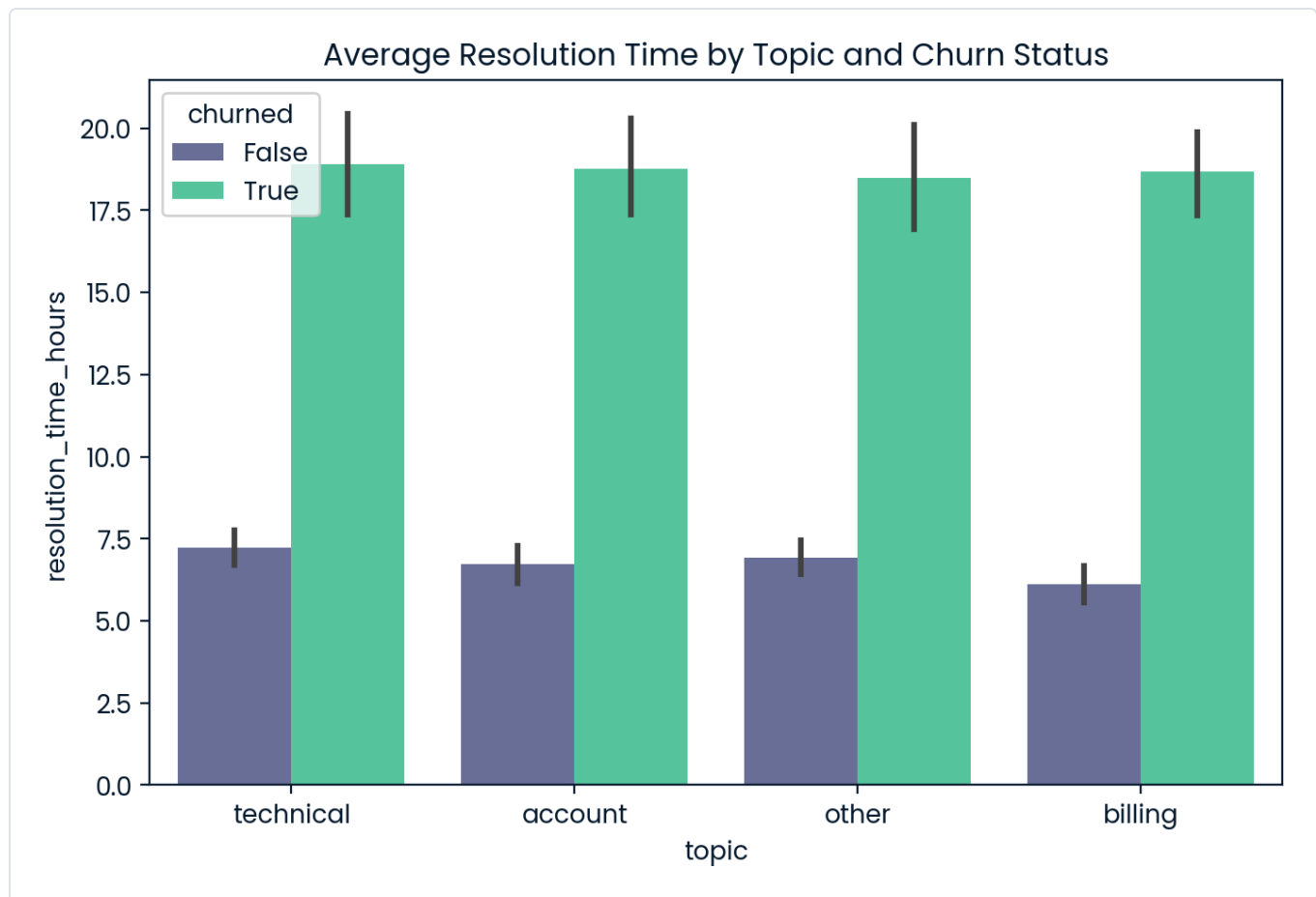


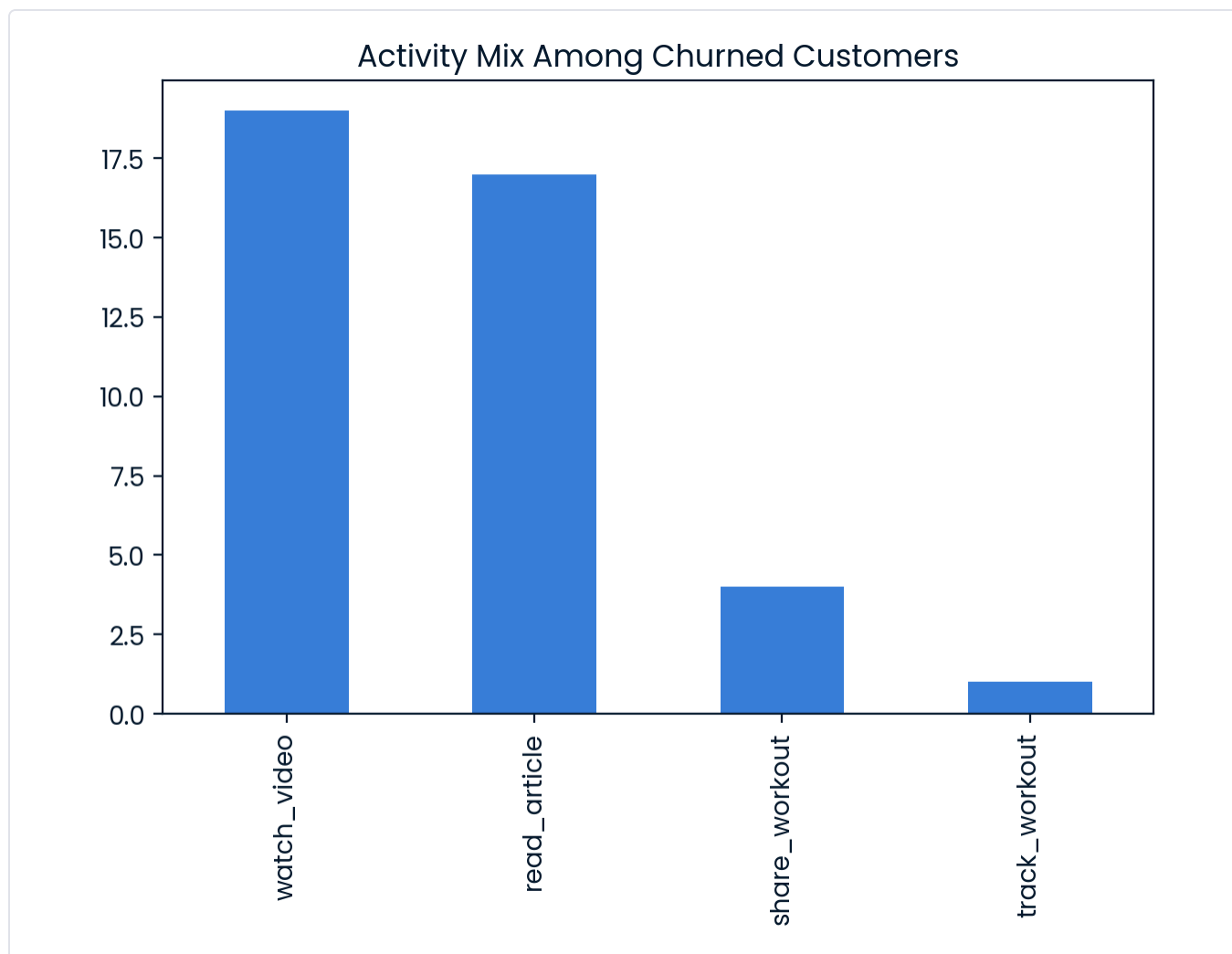
Distribution of Support Ticket Resolution Times





This relationship holds even after controlling for engagement: among customers with zero activity, those who churned still waited far longer for support resolution (18.7 hrs) than those who stayed (6.1 hrs) — nearly identical to the gap seen across the full customer base. This confirms resolution time is an independent driver of churn, not simply a byproduct of low engagement.





Customers with zero recorded product activity churn at **53.9%** — over 10x the rate of highly engaged customers (**4.2%**), a far wider gap than any split by plan type (22.4%–41.0%). Among ticketed customers, churned customers waited **18.7 hrs** on average for resolution vs **6.7 hrs** for retained ones. Engagement, not plan tier or ticket volume, is the clearest churn signal in this data.

Metric to Monitor: Customer Engagement Rate

Definition: % of customers who log at least one product activity event (video, article, workout tracked/shared) in a period. Engagement is a *leading* indicator, unlike churn itself which only shows up after cancellation.

How the business should use it: Track this rate monthly, broken out by plan tier. Because it's a leading indicator, a declining trend is an early warning that should trigger a re-engagement campaign (in-app nudges, email outreach) before those customers reach the point of cancelling — rather than finding out about the problem only after churn has already happened. A useful early-warning threshold would be a month-over-month drop of more than 3-5 percentage points, or any single plan tier falling meaningfully below the 61.5% overall baseline.

da... ...	↑↓	resolution_time_hours	...	↑↓
Sunday		11.466641791		
Tuesday		10.8011304348		
Wednesday		10.7892		
Saturday		10.3197321429		
Thursday		10.0469503546		
Friday		9.78		
Monday		9.7303846154		
Rows: 7				
 Expand				

Summary & Recommendations

Overall churn is 28.5%. Product engagement is the dominant driver (10x churn gap), support resolution speed is secondary.

1. Launch a re-engagement push for the 38.5% of customers with zero activity
2. Enforce a 10-hour support SLA, prioritizing billing tickets (36.4% churn)
3. Build a Free-tier upgrade path (41.0% churn, highest of any plan)
4. Fix the missing `signup_date` field and the ambiguous `customer_support.state` field
5. Route the 36 GDPR deletion requests to the compliance process